relying on the transport protocol to provide encryption and authentication, the message itself is encrypted and authenticated. For example, an API that expects requests with a JSON payload (or an efficient binary alternative) can be adapted to instead accept data that has been encrypted with an authenticated encryption algorithm, which it then manually decrypts and verifies as shown in figure 12.7. This ensures that an API request encrypted by the original client can only be decrypted by the destination API, no matter how many different network protocols are used to transport the request from the client to its destination.

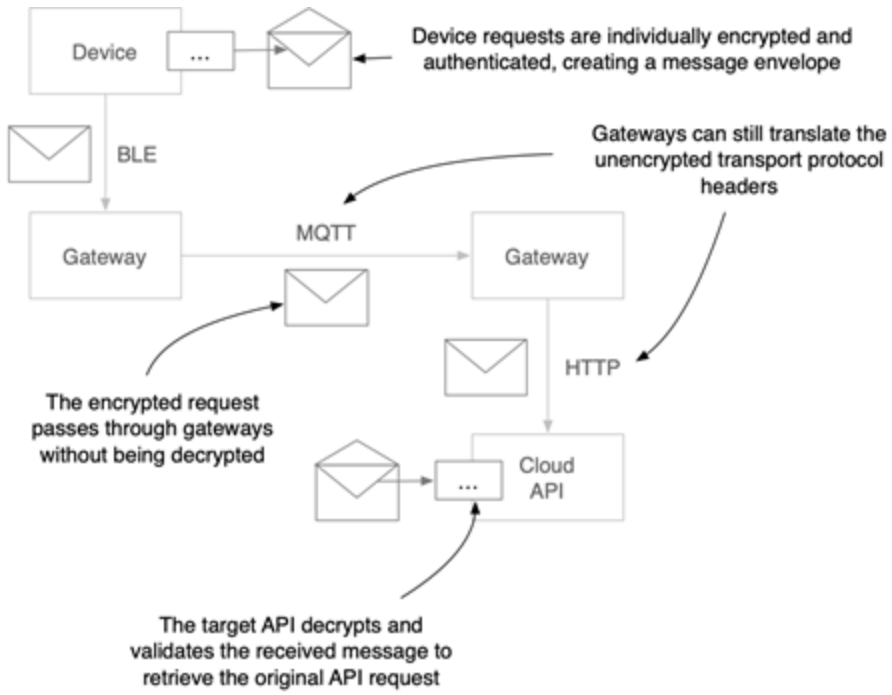


Figure 12.7 In end-to-end security, API requests are individually encrypted and authenticated by the client device. These encrypted requests can then traverse multiple transport protocols without being decrypted. The API can then decrypt the request and